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import pandas as pd

def candidate_elimination_algorithm(filename):
    # Load dataset
    data = pd.read_csv(filename)
    concepts = data.iloc[:, :-1].values
    target = data.iloc[:, -1].values

    # Initialize specific (S) and general (G) hypotheses
    specific_h = concepts[0].copy()
    general_h = [{"?" for _ in range(len(specific_h))] for _ in range(len(specific_h))

    print("Initial Specific Hypothesis:", specific_h)
    print("Initial General Hypotheses:", general_h)

    # Iterate through training examples
    for i, h in enumerate(concepts):
        if target[i] == "Yes": # Positive example
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    specific_h[x] = "?"
                    general_h[x][x] = "?"
        else: # Negative example
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    general_h[x][x] = specific_h[x]
            else:
                general_h[x][x] = "?"

        print(f"\nTraining Example {i+1}: {h}, Target: {target[i]}")
        print("Specific Hypothesis:", specific_h)
        print("General Hypotheses:", general_h)

    # Remove redundant hypotheses
    general_h = [g for g in general_h if any(attr != "?" for attr in g)]

    print("\nFinal Specific Hypothesis:", specific_h)
    print("Final General Hypotheses:", general_h)
```

```
# -----  
# DEMO RUN  
# -----  
  
# Create a small dataset inline for testing  
demo_data = """Sky,AirTemp,Humidity,Wind,Water,Forecast,EnjoySport  
Sunny,Warm,Normal,Strong,Warm,Same,Yes  
Sunny,Warm,High,Strong,Warm,Same,Yes  
Rainy,Cold,High,Strong,Warm,Change,No  
Sunny,Warm,High,Strong,Cool,Change,Yes  
"""  
  
# Save demo dataset to CSV  
with open("training_data.csv", "w") as f:  
    f.write(demo_data)  
  
# Run algorithm on demo dataset  
candidate_elimination_algorithm("training_data.csv")
```

Initial Specific Hypothesis: ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']  
Initial General Hypotheses: [['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

Training Example 1: ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same'], Target: Yes  
Specific Hypothesis: ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']  
General Hypotheses: [['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

Training Example 2: ['Sunny' 'Warm' 'High' 'Strong' 'Warm' 'Same'], Target: Yes  
Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']  
General Hypotheses: [['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

Training Example 3: ['Rainy' 'Cold' 'High' 'Strong' 'Warm' 'Change'], Target: No  
Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']  
General Hypotheses: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

Training Example 4: ['Sunny' 'Warm' 'High' 'Strong' 'Cool' 'Change'], Target: Yes  
Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' '?' '?']  
General Hypotheses: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

Final Specific Hypothesis: ['Sunny' 'Warm' '?' 'Strong' '?' '?']  
Final General Hypotheses: [['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?']]

